

Licensing



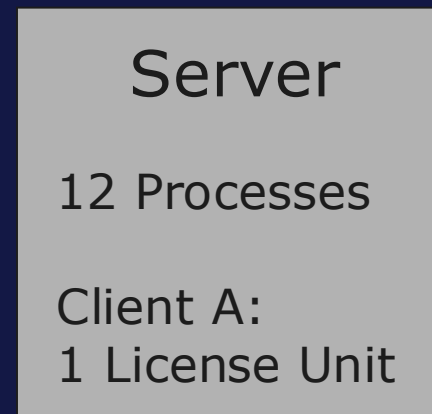
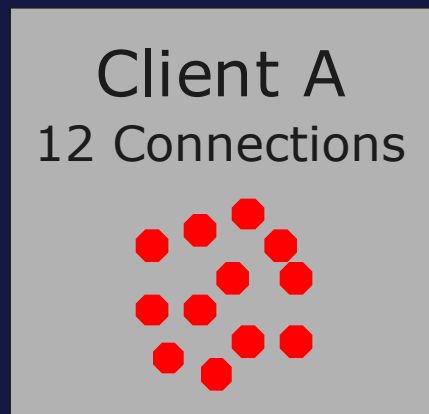
Overview

- License units.
- License processes.
- License configuration.
- License utilities.



License Units

- All licenses define license units.
 - Maximum number of allowed concurrent users.
- Each user may connect up to 12 times and use one license unit.
 - For users with more than 12 connections, each connection will consume one license slot.



License Key

- InterSystems provides license key.
 - iris.key.
- Install in every configuration.

License Capacity	InterSystems IRIS 2019.1 Enterprise - Concurrent Users for x86-64 (Microsoft Windows):50, Non-Production Instance, Natural Lang
Customer Name	ISC Learning Services - Instructor Key
Order Number	202002669
Expiration Date	1/31/2021
Authorization Key	4125500005000005000000000000F66DD90C176024F3C002 Product=Enterprise License Type=Concurrent User Server=Multi Platform=X86-64(Windows) License Units=50 Licensed Cores=0 Authorized Cores=unlimited Extended Features=024F3C00 - Interoperability

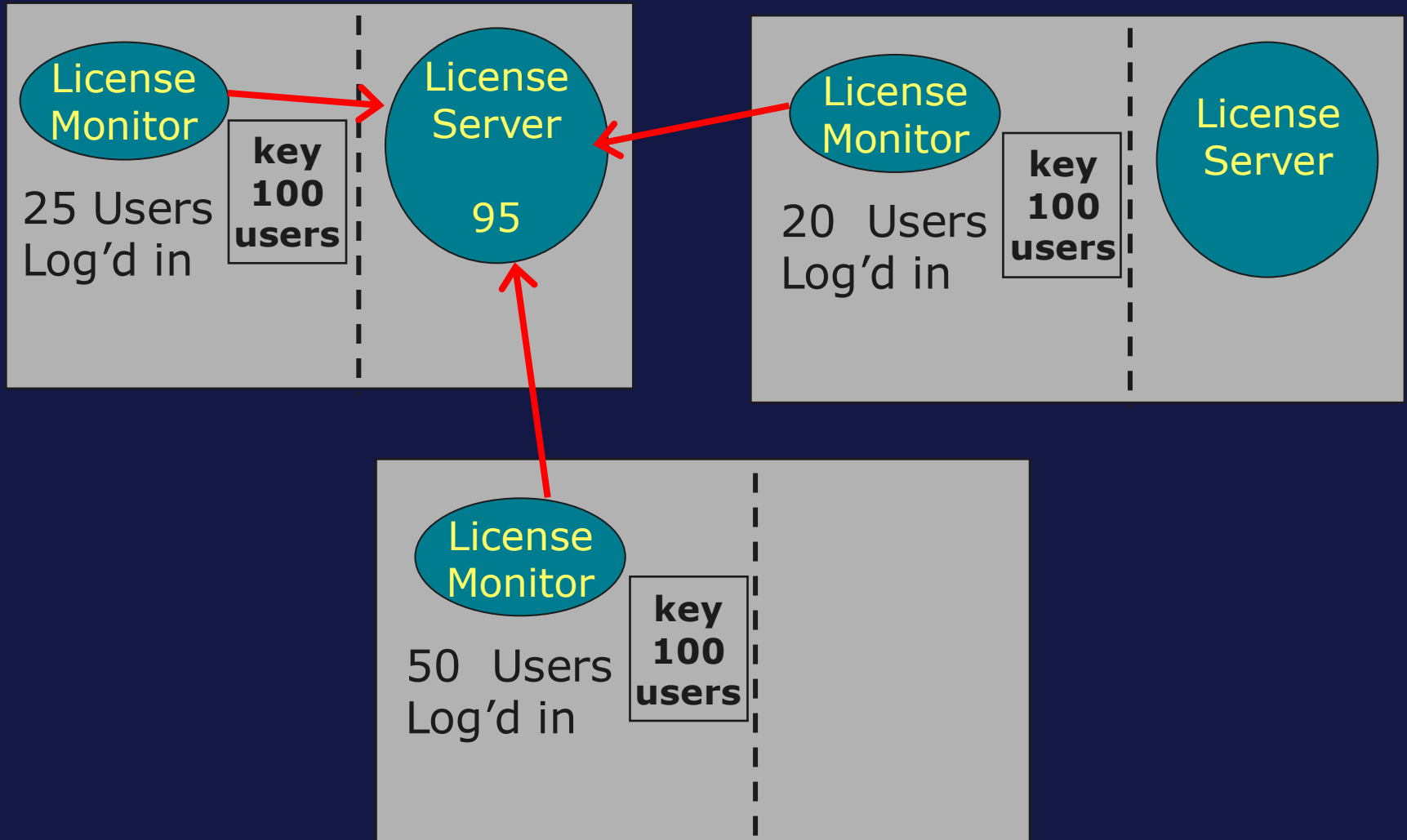


Activating License Key

- Copy key file to *InstallDirectory/mgr*.
- To activate, do one of the following:
 - System Administration > Licensing > License Key > Activate New Key.
 - `do $SYSTEM.License.Upgrade()`
 - Restart system.



Licensing Example



License Server Process

- Maintains global count of active users.
- Communicates with each License Monitor.
- Can configure multiple License Servers.
 - Fails over automatically.
- OS-level process only, identified as licmanager.
 - At most one per machine, regardless of number of instances on machine.
 - Does not require instance running on machine.
 - Stops when not active License Server for any configuration.



License Monitor

- InterSystems IRIS process.
- Provides local license limit enforcement during process startup and shutdown.
 - License Server tracks global license count.
- Periodically sends updates to License Server.
- Process name is LMFMON.



License Server Configuration

- System Administration > Licensing > License Server.
 - Descriptive name.
 - Host name or IP address.
 - Port number.
 - Default: 4002.
- All configurations sharing license key must use same set of servers.
 - Can use same port number on all servers.

The following is a list of license servers:

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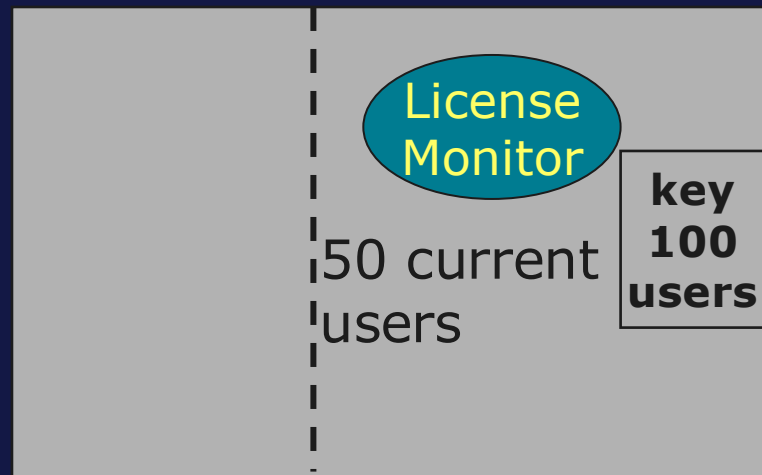
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	Name	Host Name/IP Address	Port	KeyDirectory	
	SERVER1	10.1.0.109	4002		Delete
	SERVER2	10.1.0.114	4002		Delete

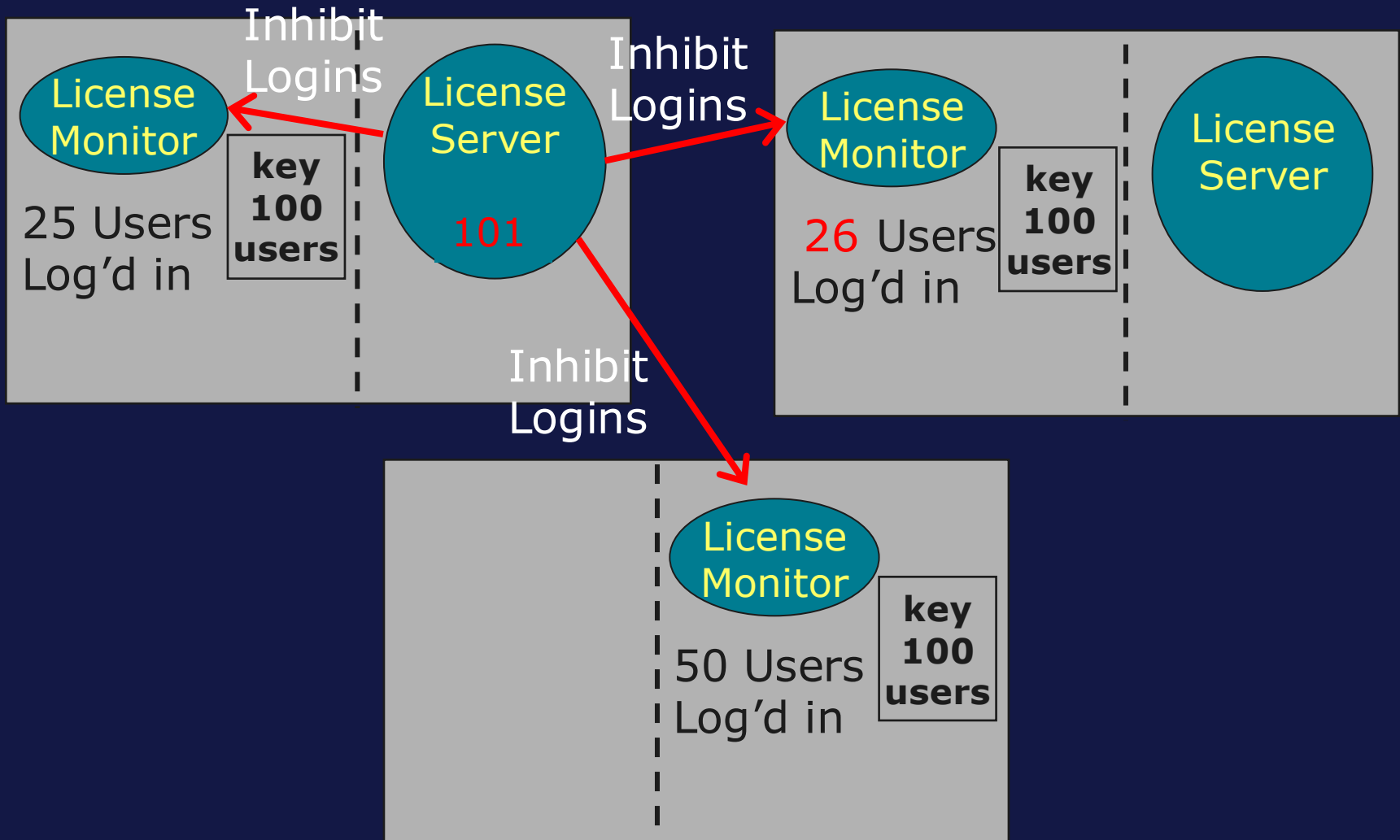


Enforcing License Limit

- License Monitor provides local license limit enforcement during process startup and shutdown.
 - Key file installed on local machine.
 - Internal tables track current license use.
 - `<License Limit Exceeded>` message given when over limit.



Exceeding License Limit



License Unit Recovery

- License units recovered when:
 - User logs out or disconnects.
 - Instance sharing license key goes down.
 - Process terminates.
 - If terminated at OS level, recovered by background process that frees license unit and some other resources held by terminated processes.



Portal License Usage Information

- System Operation > License Usage.

- **Summary**

- Usage by Process
- Usage by User
- Distributed License Usage

Current license activity summary:

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	License Unit Use	Local	Distributed
	Current License Units Used	2	2
	Maximum License Units Used	2	2
	License Units Authorized	50	50
	Current Connections	2	2
	Maximum Connections	2	2



Reviewing License Information

- License use tracked:
 - In individual configurations.
 - In License Server.
- Can review license information using `$SYSTEM.License.Method()`.
 - Examples:
 - `do $SYSTEM.License.ShowSummary()`
 - `do $SYSTEM.License.ShowCounts()`
 - `do $SYSTEM.License.ShowServer()`
 - Refer to *Class Reference Information* for complete listing.



No Licenses and Terminal

- For troubleshooting, can start Terminal session on IRIS server when no license units available:

```
irissession InstanceName -B
```



Suggested Readings

- *System Administration Guide*
 - *Managing InterSystems IRIS Licensing.*



Summary

- What are the key points for this module?

